



Impact of inter-seasonal solar variability on the association of lower troposphere and cold point tropopause in the tropics: Observations using RO data from COSMIC

V. Kumar^a, S.K. Dhaka^{a,*}, Shu-Peng Ho^b, Narendra Singh^c, Vir Singh^d, K.K. Reddy^e, H.-Y. Chun^f

^a Radio and Atmospheric Physics Lab., Rajdhani College, University of Delhi, India

^b COSMIC Program Office, University Corporation for Atmospheric Research, Boulder, CO, USA

^c Aryabhata Research Institute of observational Sciences (ARIES), Nainital, India

^d Department of Physics, Indian Institute of Technology, Roorkee, India

^e Department of Physics, Yogi Vemana University, India

^f Department of Atmospheric Science, Yonsei University, Seoul, South Korea

ABSTRACT

Association of lower tropospheric variations with the cold point tropopause (CPT) is examined on inter-seasonal basis over the tropical region (30°N–30°S) during 2007–2010 using COSMIC/FORMOST-3 Radio Occultation (RO) data. Temperature analyses for this association are shown over different regions of the globe having contrast topography namely over Western Pacific sector, Indian sector, and African sector. Correlation coefficient (r), taken as a measurement of association, show specific longitudinal differences between the lower troposphere (from 1 km to 5 km height) and the CPT. The northern and southern hemispheres show contrast coupling of temperature variation between lower tropospheric region and the CPT. Land and ocean effects are found to contribute in a different way to the correlation coefficient. Analyses show symmetrical structure of ' r ' on both sides of the equator over the African region, as data include mostly land region on both side of equator. Data represent positive correlation ($r \sim 0.5$) over 15°–20° latitudes on either side of the equator over the African region, suggesting strong hold of the inter-seasonal variation of solar diabatic heating influence over the tropic of Cancer and tropic of Capricorn. On the other hand, there is a contrast behaviour over the Indian region, ' r ' is nearly negative (~ -1.0) each year in the southern hemisphere (SH) and positive (~ 0.4) in the northern hemisphere (NH) with a maxima near tropic of Cancer. Western Pacific region is found to display a linear increase in ' r ' from negative (~ -1.0) in SH to positive (~ 0.8) in NH. In general, ' r ' (positive) maximizes over the land region around 15°–20° latitudes, suggesting a control of in phase inter-seasonal solar heating on the coupling of boundary layer/lower troposphere and CPT region, whereas it turns negative over water body. Analyses suggest that variabilities in CPT over different regions of globe show significant inter-seasonal association with the lower troposphere. Thus CPT variabilities are not only governed by QBO, ENSO, gravity waves and Kelvin wave system as reported in earlier studies but also considerably affected by inter-seasonal changes taking place in the lower troposphere.

1. Introduction

Thermal state of the Earth's atmosphere is the most important property controlling its structure and dynamics. In the recent years, there has been a growing interest to understand the variability in the tropopause region, as it responds to different forcing on a varied time scale. Tropopause is colder, higher and sharper during boreal winter than during boreal summer. In addition, this region exhibits long time scale signal in temperature that may have a link to the climate change

(Dhaka et al., 2010). Annual variability of this region is influenced by the large-scale motion, such as QBO from the top side and on seasonal scale from the lower side (Kumar et al., 2014). Meridional connection of polar region especially due to SSW event with that of the tropical region is also of concern that can alter the characteristic features of tropical tropopause region for several weeks (Dhaka et al., 2015). In the past, there have been various studies which focused on the tropopause dynamics (Highwood and Hoskins, 1998; Kerr-Munslow and Norton, 2006; Ueyama and Wallace, 2010; Chun et al., 2007, 2011; Randel and

* Corresponding author at: Department of Physics and Electronics, Rajdhani College, University of Delhi, Raja Garden, New Delhi 110015, India.
E-mail address: skdhaka@rajdhani.du.ac.in (S.K. Dhaka).

Hensen, 2013; Kumar et al., 2014). Annual variation of tropopause is usually discussed with the significant role of vertically propagating Rossby and gravity waves (Garcia, 1987; Gray and Dunkerton, 1990; Haynes et al., 1991; Yulaeva et al., 1994; Holton et al., 1995). However, the Brewer–Dobson circulation (BDC), in addition to QBO, is the important factor affecting annual variation of tropopause region (Ueyama and Wallace, 2010; Chun et al., 2011; Randel et al., 2003; Kumar et al., 2014). Other factors such as ENSO (on large time scale) and MJO (intra-seasonal scale) also influence the tropopause structure (Scherllin-Pirscher et al., 2012; Zeng et al., 2012; Dhaka et al., 2007, 2010). Kim and Son (2012) revealed that inter-seasonal cycle of tropopause temperature is modulated by tropospheric circulations. But on intra-seasonal time scales, the variability of tropopause properties is primarily controlled by Kelvin waves, with a non-negligible contribution by Madden-Julian oscillation convection.

Taking into account the various forcing that may influence the tropopause structure, the most important aspect still needs to be examined viz. how closely the variability of tropopause and lower atmospheric dynamics are coupled. This has not been explored in detail during the past studies. Upper and lower tropospheric regions are connected through convection and deep convection heat transfer. Chun et al. (2007) (and references therein) have described the inter-seasonal solar influence on earth's surface changes its temperature and plays an important role in terms of convective sources and deep convective heating (DCH). Hence, it becomes essential to investigate how the coupling between tropopause and the lower troposphere (1 to 5 km) including the boundary layer processes varies with inter-seasonal solar influence under different topographical conditions especially in the tropical belt.

The cold point temperature (CPT) is the lowest temperature in the tropopause region and represents the upper boundary of tropical tropopause layer (Gettelman and de F. Forster, 2002). The tropical CPT marks the division between convection and the radiative control of temperature profiles (Held, 1982). Balance between radiative and large-scale advective sources of energy cannot be achieved until some distance above the CPT, when net radiative heating prevails near the tropopause (Sherwood, 2000), if thin cirrus clouds contribute to the heating (Jensen et al., 1996). Observed temperature trends and inter-annual variations near the tropical tropopause suggest that temperatures up to the CPT are controlled by the troposphere (Sherwood et al., 2003).

Sun and Oort (1995) found an almost constant vertical temperature correlation of ~ 0.7 with variations in specific humidity using radiosonde data over a period of 1963–1989. Lanzante (1996) commented on the correlation coefficient reported by Sun and Oort (1995), giving some limitations and reliability issues in data especially in upper troposphere. Later, Bauer et al. (2002) found that spatial averages of the local temperature-humidity correlations are much weaker, and show more height dependence than the correlations of the spatially averaged quantities for both model and observed data. Hence, it is brought out that correlation coefficient reported by Sun and Oort (1995) due to the poor spatial coverage resulted into a significant bias.

Wu et al. (2006) have attempted using satellite, radiosonde measurements, and General Circulation Models (GCMs) to investigate the vertical correlations of temperature variations at 925 hPa with the atmospheric boundary layer, free troposphere, and lower stratosphere over a widely averaged tropical region (20°N–20°S). Their analysis has suggested that surface temperature with upper troposphere is positively correlated (~ 0.4), while tropopause region and lower stratosphere are negatively correlated. This study did not reveal the actual coupling of lower troposphere with the upper troposphere on both side of equator, and the hemispherical differences could not be discerned. Note that the Atmospheric Infrared Sounder (AIRS) retrieval temperature product (Olsen et al., 2005), which was onboard NASA Aqua satellite, were used for the analysis. Due to the vertical broad vertical resolution (~ 3 km in the troposphere) and long correlation length in the background

covariance matrix, their temperature retrievals may be highly depending on the variation of degree of freedom of signals of AIRS temperature channels which are highly correlated to surface and troposphere temperature contact. In addition, although using cloud clearing algorithm (Olsen et al., 2005), AIRS data are still very limited to clear skies. AIRS samples over cloud dominated regions (strong convection regions) are mainly collected over cloud edge where cloud clearing algorithm can still apply.

Unlike infrared sounders (for example, AIRS) which are limited to clear skies, Global Positioning System (GPS) Radio Occultation (RO) data provide all-weather high vertical resolution refractivity profiles. With a very high vertical resolution (from ~ 100 m near the surface to ~ 1.5 km at 40 km) and a very short correlation length, RO derived variables are more suitable for studying the correlation between the lower troposphere (LT) and the CPT than that from AIRS retrievals (i.e., Wu et al., 2006). In this study, we will use Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC)/Formosat Satellite Mission (COSMIC/FORMOSAT-3, hereafter referred as COSMIC) RO measurements to investigate the correlation between the lower troposphere (from 1 km to 5 km height) and the CPT. Because COSMIC data are uniformly spread over land and oceans which provides an opportunity to examine the association of lower and upper troposphere categorically on each side of the equator. Moreover, results can be obtained in different latitudes bands and seasons. COSMIC observations widely supported several issues of atmospheric processes, including an identifying the atmospheric boundary layer (e.g. Ratnam and Basha, 2010; Ho et al., 2015). The top of the atmospheric boundary layer (ABL) height shows an inter-seasonal and land/ocean variability, with maximum height in tropical region (~ 2.0 km). In general, NH summer shows higher values of ABL over continents in the tropical belt, except over Indonesian region, where during winter maximum height was attained. This is a useful information to differentiate between ABL and lower troposphere region and also important to differentiate the region above ABL from the lower free troposphere.

Current study is useful for the better understanding of association of lower and upper troposphere, including the relative influence of lower troposphere below and above ABL on CPT.

In this study, we present the correlation between variabilities of lower troposphere temperature and CPT at each 5° non-overlapping latitude band, over entire tropical region (30°S to 30°N). To examine the land and ocean effects separately on the correlation, results are classified over three different longitudinal sectors namely (a) Indian (60°E–120°E), (b) African (10°E–40°E), and (c) Western Pacific (120°W–180°W) region.

Temperature data, on a fine temporal and spatial scale, from COSMIC provides an excellent opportunity to explore the control of inter-seasonal solar influence on the coupling of lower troposphere and CPT region. The present study will demonstrate the spatial distribution of correlation coefficient using an impressive temperature data from COSMIC satellite.

2. Data and methodology

Radio occultation (RO) technique has emerged as a powerful sounding for the global atmosphere with high precision, accuracy, and vertical resolution in all-weather conditions over both land and the ocean. RO technology offers great potential for atmospheric and ionospheric research, improvement of numerical weather forecasts, space weather monitoring and climate change detection (Kursinski et al., 1997; Anthes et al., 2000; Hajj et al., 2000; Kuo et al., 2000). COSMIC mission is a collaborative project of the National Space Organization (NSPO), Taiwan and the University Corporation for Atmospheric Research (UCAR), United States. It is the first RO mission to use a constellation of six satellites in the low earth orbit (LEO). The observed number of Occultation are about 2500 per day with approximately the uniform distribution in time and space (Ho et al., 2009). COSMIC data

are of better quality than those from the previous missions (CHAMP and SAC-C) that penetrate much closer to the Earth's surface, and 70%–90% of the soundings reach to within 1 km of the surface on a global basis (Anthes et al., 2008; Ho et al., 2013, 2010a, 2010b). It has been demonstrated that COSMIC data in the neutral atmosphere can be used to study modes of variability above, within, and below clouds (Biondi et al., 2012, 2013; Teng et al., 2013; Huang et al., 2013; Scherllin-Pirscher et al., 2012; Zeng et al., 2012; Mears et al., 2012, 2013; Rieckh et al., 2017) and atmospheric temperature and refractivity trends in the lower stratosphere (Ho et al., 2010a, 2010b, 2012). More details can be found in Anthes et al. (2008, 2000) and Ho et al. (2013). Due to the deeper penetration, the lower troposphere COSMIC instruments can observe the structure of the tropical atmospheric boundary layer (Anthes et al., 2008). COSMIC is the first RO mission that demonstrated a consistent ability to profile the vertical structure of the atmospheric boundary layer (Anthes et al., 2008; Ho et al., 2015).

COSMIC Level-2 wet profiles of temperature have been utilized in this study (available at www.cosmic.ucar.edu). With open-loop tracking, COSMIC receivers are able to track RO signals with sufficient signal-to-noise ratio (SNR). As a result, RO profiles generally reach lower than 2-km altitude (Sokolovskiy, 2001). In the neutral atmosphere, the atmospheric refractivity N is a function of the pressure P , temperature T , water vapor pressure P_w , through the following relationship (Kursinski et al., 1997):

$$N = 77.6 \frac{P}{T} + 3.73 \times 10^5 \frac{P_w}{T^2} \quad (1)$$

where P is pressure in hPa, T is temperature in K, P_w is water vapor pressure in hPa. A 1D-var algorithm (<http://cosmic-io.cosmic.ucar.edu/cdaac/doc/documents/1dvar.pdf>) is used to derive optimal temperature and water vapor profiles where temperatures and water vapor profiles from the ERA-Interim reanalysis are used as a priori estimates (Zeng et al., 2012). As demonstrated by Ho et al. (2007, 2010a, 2010b) that COSMIC refractivity data are very sensitive to water vapor variation and less sensitive to temperature. Therefore, COSMIC temperature data used in this study are primarily close to (but not equal to) a priori (ERA-Interim reanalysis) which include not only RO data but also Advanced Microwave Sounding Units (AMSU) for both clear and cloudy skies and infrared data for clear sky.

Anthes et al. (2008) and Ao et al. (2003) illustrated that RO data may contain negative refractivity bias due to the super-refraction. The super-refraction usually occurs over regions with a relatively cold sea surface temperature in the subtropics and mid-latitudes (i.e., the Southeastern Pacific, Northeastern Pacific, and Southeastern Atlantic, see Ho et al., 2015). Because the refractivity (N) is more sensitive to water vapor variation than that to temperature (Ho et al., 2007, 2010a, 2010b), the water vapor retrievals may be biased over “super-refraction” regions, but the temperature retrievals may still be not affected. These are demonstrated in comparison of RO temperature profiles with high resolution radiosonde data over VAMOS Ocean-Cloud-Atmosphere-Land Study Regional Experiment (not shown).

The vertical resolution of RO derived humidity profile is from ~100 to 200 m in the lower troposphere and ~1.4 km in the stratosphere. Such a remarkable near-real time data have been available for the first time with satellite observations, having a wide spatial coverage with high accuracy (average accuracy ~0.1 K) and high vertical resolution (Ho et al., 2009). COSMIC data have been validated with NCEP, JRA-25, and UK Met Office data sets (Kishore et al., 2009). In Table 1, we have shown the number of occultations derived from COSMIC between 30°N and 30°S over the period of 2007–2010.

COSMIC temperature profiles from January 2007 to December 2010 are used to examine the association through correlation coefficient between lower troposphere and CPT region. During each year, all the occultations are averaged monthly, at each 5° non-overlap latitude band, between 30°N–30°S ranging over a total of 12 latitude bands. Each latitude band generally contains ~1400 occultation points in a

Table 1

Number of occultations observed during 2007–2010 from COSMIC in tropics (30°N–30°S).

Years	Number of occultations
2007	2,14,796
2008	2,15,092
2009	2,10,814
2010	1,61,457
Total	8,02,159

month. Then, monthly zonal mean temperature profiles for each latitude band are constructed at 100 m height intervals between surface and 40 km. Using these monthly profiles, the correlation coefficient of temperature between different tropospheric height levels and the CPT are examined on the basis of monthly variation during a year. This correlation coefficient is calculated separately for each year during 2007 to 2010. Same methodology for the correlation coefficient is followed over Indian sector (60°E–120°E), African region (10°E–40°E), and Western Pacific (120°W–180°W) sectors.

3. Results and discussion

Temporal and spatial variability of CPT in the tropical region is an important aspect, which illustrates how the processes are taking place and operate in tropical troposphere and lower stratosphere (Kim and Son, 2012; Kumar et al., 2014). As cited above, there are some known atmospheric processes that act on different time scales, which in turn illustrate partly the cause of CPT variability. For instance, different descending phases of QBO can explain the long term anomalies in temperature at CPT level; on the other hand on a few days basis some wave structures were observed in CPT which were caused by Kelvin wave activity (Dhaka et al., 2007; Panwar et al., 2012; Kumar et al., 2014). Systematic inter-seasonal changes in CPT still needs proper examination to develop a clear understanding about the control forcing scales particularly from the lower tropospheric side.

One of the most important parameters is the strong inter-seasonal solar heating process, for instance, during NH summer tropical atmosphere develops large convective sources resulting precipitation and hence deep convective heating (Chun et al., 2007). This can influence the association of temperature significantly between the lower height and the CPT region. Therefore, it is essential to investigate the inter-seasonal variation of CPT with the inter-seasonal variation of temperature at different height levels in the lower troposphere. This aspect is helpful to explore the role of inter-seasonal change in temperature at different heights levels for CPT dynamics.

We calculated correlations of temperature data between lower tropospheric different height levels and the CPT region on daily and monthly basis data during a year; the zonally averaged features were presented between 30°S and 30°N. Vertical correlations are shown in the height range of 1 km to 20 km. Similarities and differences are examined in both hemispheres; correlations are shown from equator to 30°N and 30°S, respectively. These are computed using daily as well as monthly mean data so that dominance of the process on a small scale (less than a month) as well as on an inter-seasonal scale can be distinguished in the tropical atmosphere. In all cases, correlations are observed to be statistically significant at 99% confidence level.

Fig. 1 shows vertical correlation coefficient of the CPT with the different levels from 1 km to 20 km height at every 100 m using daily (left panel) and monthly mean (right panel) variation of temperatures from 1 January 2009 to 31 December 2009. Vertical correlations are representative of zonally averaged data covering large tropical region between 30°S and 30°N. In general, correlations are positive in daily as well as monthly data, nonetheless, the values are higher (~0.75) with monthly data suggesting better association with inter-seasonal changes in comparison to daily data. Fluctuations of periods less than a month

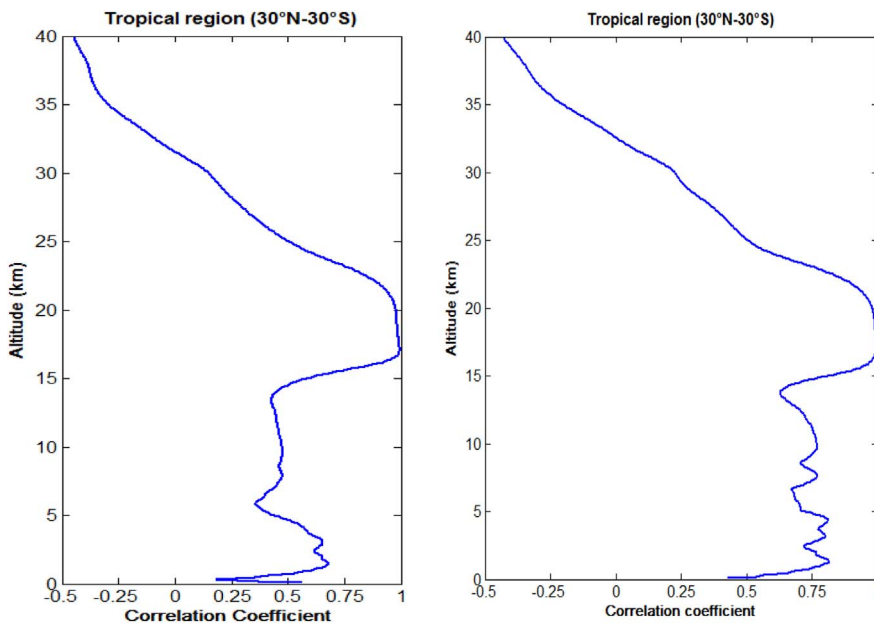


Fig. 1. Altitude profile represents the correlation coefficient of CPT with temperatures in the tropical region (30°N–30°S) at different levels starting from lower troposphere to 40 km altitude at each 100 m using daily (left panel) and monthly (right panel) variation of temperature data during year 2009. Correlations are shown at 99% confidence interval. Above 20 km altitude, inter-seasonal variability show less influence while features such as QBO dominates hence a gradual decrease in “r” is seen from 20 km to 40 km altitudes due to downward phase propagation of oscillations.

are quite common and prominent when we use daily data in the troposphere. Correlation coefficient is about 0.4 in the middle and upper troposphere with an indication of slightly large values at lower troposphere. These values are consistent with the result given by Wu et al. (2006) in the tropical troposphere, though, tropical averaged values in their analyses were representing from 20°S to 20°N latitudes.

Correlations shown in Fig. 1 exhibit similar vertical tendency in both daily and monthly data, except that large vertical correlation with monthly variation. Similar features of correlation coefficient were also observed during other years from 2008 to 2010 (not shown here). In general, the CPT was positively correlated in the troposphere and lower stratosphere in tropical region. However, in the range of 20 km to 40 km altitude, inter-seasonal variability showed less influence since large period oscillations such as QBO dominates with significant amplitudes. Hence a gradual decrease in “r” is seen upwards due to downward phase propagation of oscillations in this thick layer.

There is a large spatial difference of topography between NH and SH. We have further divided the region of study for vertical correlation from equator to 30°N and 30°S, respectively. To examine the hemispherical differences we used monthly data on both side of equator and vertical correlations are shown in Fig. 2 for the year 2009. Left panel represents correlation coefficient in SH (0–30°S) and right panel in NH (0–30°N). It is clearly seen from Fig. 2 that both hemispheres show contrast correlation coefficient in tropical tropospheric region up to 10 km altitudes. The northern hemispheric CPT region is positively correlated (~ 0.85) with lower tropospheric region, while in the southern hemisphere a negative correlation (~ -0.75) is prominent. Note that there is contrast behaviour in the vertical correlations. Similar to Fig. 1, a gradual decrease in “r” is seen from 20 km to 40 km altitudes due to a consistence downward phase propagation of QBO oscillations.

Latitude-height section covering from 30°S to 30°N, zonally averaged features of vertical correlations in the range of 1 km–20 km height is shown in Annexure A during 2007–2010. Annexure shows a clear difference of tropical troposphere in northern and southern hemispheres. Inter-annual variation, in Annexure A, is illustrated. During El Niño and La Nina years correlations are seen varying significantly in the lower and upper troposphere. These features suggest the key role of the topography on both sides of the equator. Land and ocean body shows contrast behaviour in responding the forcing from lower troposphere to upper troposphere. This peculiar behaviour of tropical troposphere in two different hemispheres was obscured when data was studied by averaging over a large tropical region and such characteristic features

in the earlier studies (Wu et al., 2006) were not revealed.

Chun et al. (2007) have illustrated Deep convection heating (DCH) covering tropical latitudes using satellite and NCEP data during 1985–1986. There might be some difference in DCH values with the period 2007–2010, although general features over longitude regions remain same. They have shown that DCH enhances near 10–20°N and 10–20°S, with a little larger values on northern hemispheric side. Their Figs. 1 and 2 have shown clearly that northern hemisphere have larger values of deep convective sources on different regions including Indian (Bay of Bengal area) region and northern pacific region etc. Contrast behaviour of vertical correlations in NH and SH are primarily believed due to DCH by the presence of large convective activities during summer in NH in comparison to SH.

In next sections, we have further classified the results over different parts of globe such as African region, Indian region, and Pacific region. This would clarify the influence of land-sea contrast by taking the opportunity, especially over African region and Pacific region.

RO data are available at a fine spatial and temporal scale over different region of globe and therefore features of correlations can be studied in narrow latitudes bands. Further correlation coefficient of lower troposphere with cold point tropopause (CPT) is examined at each 5° band in the latitude range of 30°S to 30°N during 2007 to 2010. In addition to zonally (180°W to 180°E) averaged structures, prominent features over three regions, Indian sector (60°E–120°E), African region (10°E–40°E), and (d) Western Pacific (120°W–180°W), are also shown.

3.1. Correlations between CPT and lower tropospheric region in zonal mean temperature over tropical region

Correlation coefficient on inter-seasonal basis (January to December using averaged monthly data) is examined between zonal mean CPT and lower tropospheric temperature at each 0.5 km height interval between 1.0 km to 5.0 km, which is shown in Fig. 3. The value of “r” is computed at each 5° non overlapping latitude band from 30°S to 30°N. Correlations shown are at 99% confidence level.

In Fig. 3, blue, red, yellow, and green color denotes year 2007, 2008, 2009 and 2010, respectively. Black color represents “r” over tropical region averaged from 2007 to 2010. Correlation shows large variability with latitudes between 30°N and 30°S. It is evident from Fig. 3 that correlation coefficient around equatorial region (10°N–10°S) have a negative value (~ -0.5). In SH, beyond 10°S “r” between CPT and lower troposphere turns strongly negative (~ -0.9), suggesting

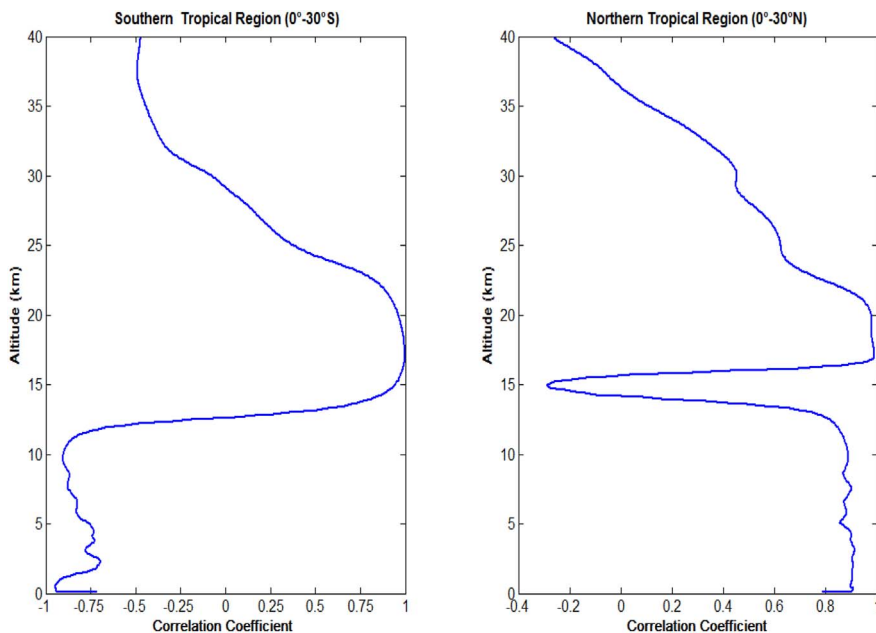


Fig. 2. Represents the correlation coefficient of CPT with temperature at different levels at each 100 m interval in southern (left panel) and northern hemispheres (right panel) using monthly averaged data. Between 20 km and 40 km altitudes, vertical correlation decreases gradually suggesting a downward phase propagation of seasonal and annual cycles along with QBO.

that these regions are anti correlated. However, in NH, beyond 10°N correlation coefficient is strongly positive (~ 0.9) near 15°N–25°N latitude band. Note that in the latitude band 25°N–30°N a sharp reduction in the values of 'r' takes place. NH and SH tropical regions show contrast features between lower atmosphere and CPT. Though the correlations between CPT and various lower tropospheric levels varies slightly they possess the same tendency i.e. negative in SH and positive in NH. Especially, 10°S–15°S latitude band shows negative correlation at 1.0 km altitude and turning positive coupling with 5.0 km altitude. Note that ABL, as observed by Ratnam and Basha (2010), exists around

2.0 km altitude, and correlations were stronger above this level suggesting a more free interaction above ABL. Ho et al. (2015) have shown Marine boundary layer heights and their longitudinal, diurnal, and inter-seasonal variability in the South eastern Pacific Using COSMIC, CALIOP, and radiosonde data. Signal of diurnal variation is about of the same order as that of annual variation in the range of 1.2 km to 1.8 km altitudes.

Inter-annual variation in vertical correlations is also quite clear in Fig. 3. Over 15°S–15°N it is maximum and positive in 2009 in comparison to other years. This feature is quite strong over equator and in

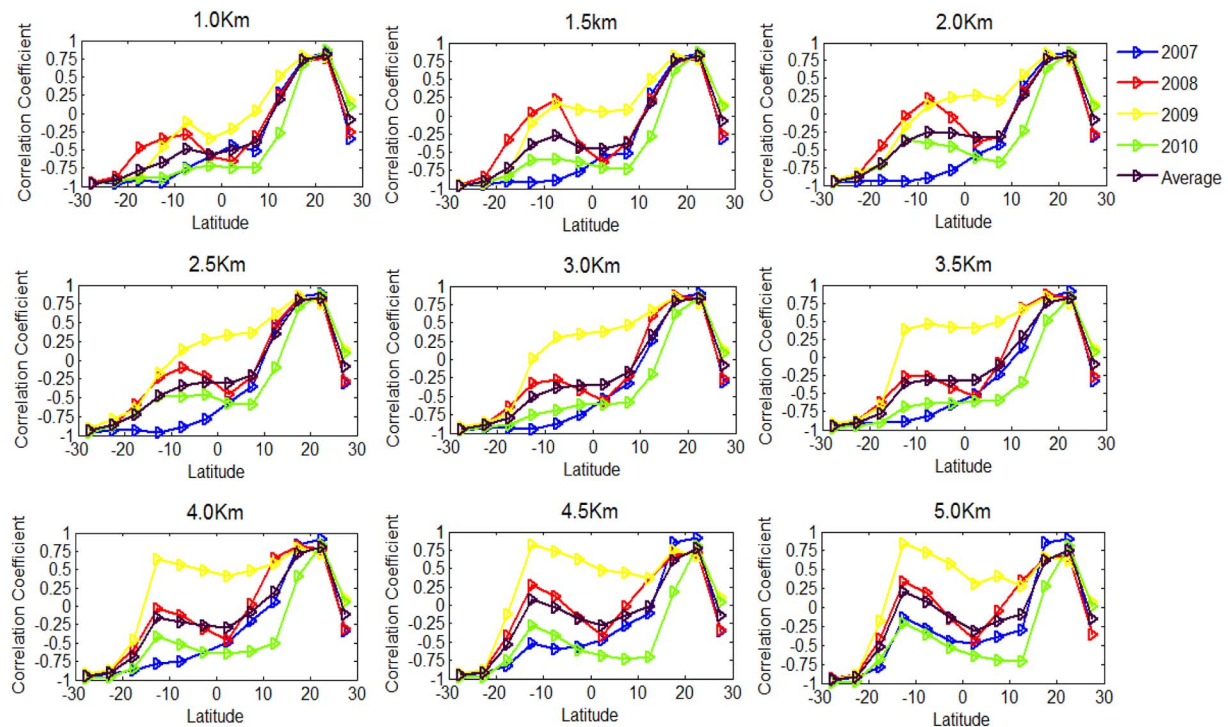


Fig. 3. Correlation coefficients (r) between CPT (using monthly data; i.e. inter-seasonal variability of zonal mean) and lower tropospheric temperature at each 0.5 km between 1.0 km to 5.0 km. Correlations are shown at each 5° non overlapping latitude band from 30°S to 30°N. Different colors denote different years (Blue, red, yellow, green color represent year 2007, 2008, 2009 and 2010 respectively, while the black color represent the average over four years). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

SH tropical region too. Analysis of global surface temperature by NASA agency shows that 2009 was the second warmest year in the modern record. In addition, 2009 is also the year of El Niño event, which Kumar et al. (2014) has shown that entire troposphere was warmer during 2009 (see Fig. 3). In our analysis, unique feature of 2009 correlations suggests a significant role of surface temperature to CPT variability. While looking at inter-annual variation, year 2008 also registered higher correlations, though the values are less than of 2009. Contrast to 2009, year 2008 was a La Nina year, and comparatively a cooling effect was observed in the entire troposphere region. El Niño has a tendency to create an opposite temperature effects near CPT than in the troposphere, i.e. El Niño cools lower stratosphere while La Nina warms it (Scherllin-Pirscher et al., 2012). These induced features of cooling and warming in the lower stratosphere adjacent to CPT seem linked with strong El Niño and La Nina periods and have positive correlations with lower atmosphere.

Wu et al. (2006) have shown in their analysis negative correlation between surface and near tropopause level temperatures on inter-seasonal variation basis; positive correlation gradually changed to negative while moving from middle and upper troposphere region to the near tropopause region. Using RO data we could bring such a fine scales variability more clearly over the tropical troposphere region. We essentially presented the coupling at each 5° latitude band, and the correlations showed a systematic change from SH to NH. To examine the roles of various small and large scale phenomena such as El Niño, convection and land-ocean effect on the coupling between CPT and lower tropospheric region, the study is further focused over (a) Indian sector (60°E – 120°E), (c) African region (10°E – 40°E), and (c) Western Pacific (120°W – 180°W) region which is described in the subsequent sections.

3.2. Correlations over different longitude sectors

3.2.1. Indian region (60°E – 120°E)

Vertical correlation is examined between CPT and lower troposphere over Indian and Indonesian longitudes (60°E – 120°E), which are shown in Fig. 4. Correlations are negative in SH (between 10 and 30 S) with high values of r (~ -0.9), whereas a positive correlation (~ 0.4) is

noticed in NH peaking near 10° – 20°N . The features of correlations from 1 km to 5 km altitude is little different; below ~ 2 km values are significantly large, whereas near 3 km and above correlations are less negative in SH. Chun et al. (2007) showed largest DCH over Indian land mass and Bay of Bengal region especially during summer monsoon region. Dhaka et al. (2010) have also shown large values of convective available potential (CAPE) energy from 1980 to 2006 using radiosonde data over Kolkata (22.3°N , 88.2°E) located near coastal region of Bay of Bengal. They have shown strong inter-seasonal and inter-annual relations of CAPE with temperature at 100 hPa pressure level. Large CAPE values were not only observed during summer season but also quite significant during winter returning monsoon too. Thus, Indian region in NH is convectively very active and hence its connection with upper troposphere is quite evident on an inter-seasonal scale. Fine scale observations also support this notion. For instance, radar observations of several convective events at Gadanki (13.5°N , 79.2°E), a nearly coastal station facing Bay of Bengal, have shown growth of deep convections within a period of 1–2 h from lower troposphere to lower stratosphere through atmospheric gravity waves (Dhaka et al., 2002, 2003). These radar observations support prevailing frequent deep convection over Indian land mass and Bay of Bengal region. Convective deep heating in SH over similar longitudes sectors is less evident largely due to presence of water body. Nature of convection is different over land and ocean because of their heat capacity (Gupta et al., 2013; Kumar et al., 2013). Inter-annual variation of vertical correlation and varying correlations above ABL in SH are quite noticeable features. Interestingly, above ABL during 2009, correlations are significantly high (~ -0.9) in SH, even larger than NH, suggesting a strong impact of ENSO over Indian Ocean. This is a remarkable feature of increasing convective activities due to warm SST centered over Indian longitudes sector.

3.2.2. African region (10°E – 40°E)

Fig. 5 represents correlation over African longitudes sector (10°E – 40°E) that covers the land region in both the hemispheres. There is a symmetrical structure of ' r ' from 30°S to 30°N over the African tropical belt. Positive correlations ($r \sim 0.25$ – 0.5) over 15° – 20° latitudes on both sides of equator are observed, particularly in 2008 and 2009. Average correlation structures indicate of weak (< 0.25) or no

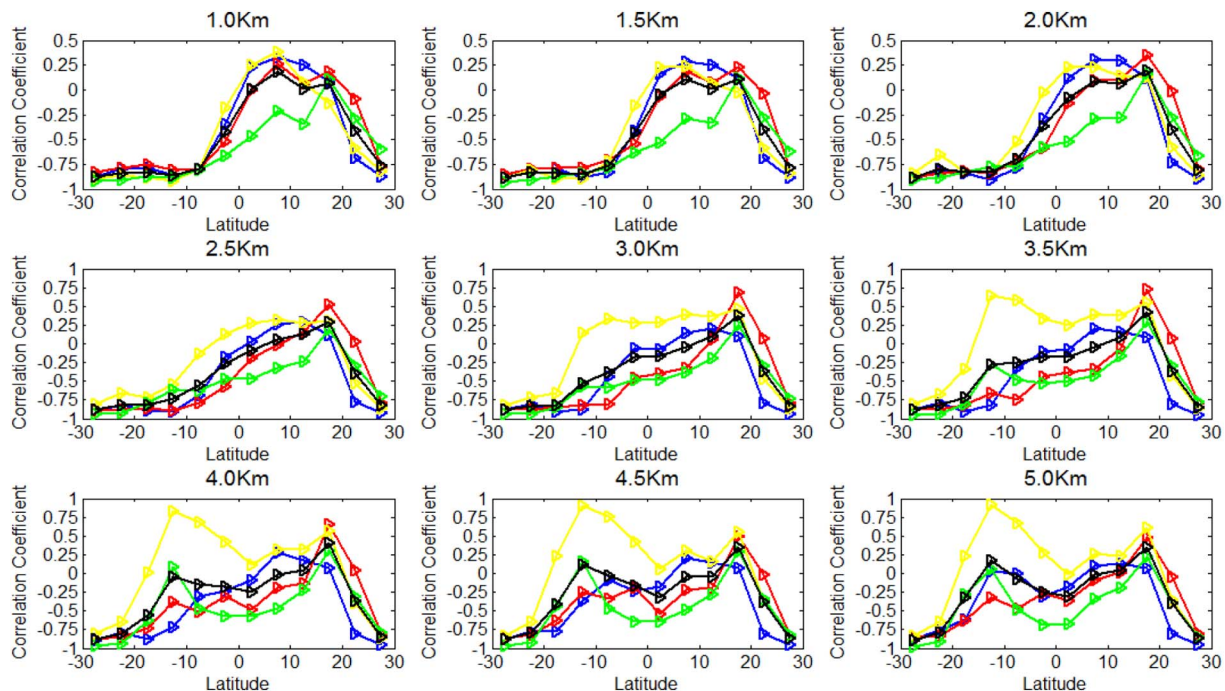


Fig. 4. Same as Fig. 3, but over Indian region (60°E – 120°E).

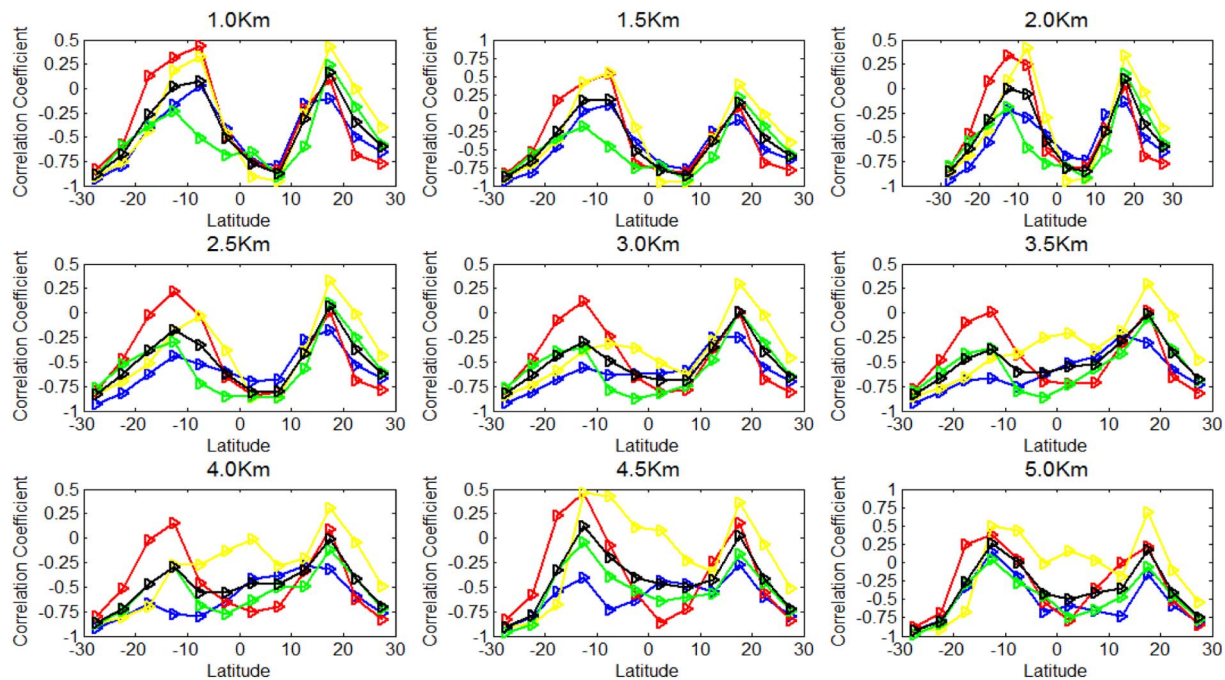


Fig. 5. Same as Fig. 3, but over African region (10°E–40°E).

correlation near 15–20 latitudes. Inter annual variation is quite evident. The equatorial region, particularly from 5°S to 10°N, shows a strong negative correlation (~ -0.75 – -0.9). Similar vertical anti-correlations are noted beyond 20°S and 20°N. DCH computed by Chun et al. (2007) over off equator regions in both hemispheres show a good agreement with the latitudinal variability of vertical correlations. Over those African regions where deep convective heating were larger, increase in correlations are consistent. Note that the DCH in African region is about half of the Indian region, and hence less correlations is expected, which is in good agreement with the correlations features. During 2009 an El Niño year, correlations increase over equatorial region noted particularly above ABL, though the impact of influencing correlations structure is less in comparison to Indian sector, where it was quite remarkable. Similarly during 2008, a La Nina year, enhancement in correlation coefficient is noted particularly in SH from 1.0 km to 5.0 km. Over Indian region, El Niño has shown a greater influence, while over African region, especially in SH, La Nina has shown larger association between lower and upper troposphere. In fact, La Nina has shown larger influence on correlation variability over Pacific region too (shown in next section), this is similar to over African region.

3.2.3. Western Pacific region (120°W–180°W)

In a recent study by Scherllin-Pirscher et al. (2012) using COSMIC RO data it has been shown that recent La Nina (2007/08) and El Niño (2009) events have given a profound impact in the entire troposphere and also in a thin layer of lower stratosphere. During La Nina, entire troposphere have a cold regime, while during El Niño it turns into a warmer state, however, lower stratospheric region near CPT shows up opposite response. Kumar et al. (2014), however, have argued that CPT characteristics are partially changed due to descending different phases of QBO. Correlation coefficient is examined over Western Pacific sector which is covered by a vast water body on both sides of equator unlike African sector. Correlation coefficient over the Pacific region (120°W–180°W) is shown in Fig. 6. Correlations, in the range from 1.0 km to 5.0 km, show almost a linear increase from strong negative (~ -1.0) to strong positive (~ 0.8) covering from 30°S to 30°N. The region does not reflect any symmetry in 'r' structure rather it showed an asymmetrical behaviour. Positive correlations are seen in the NH similar in magnitude to Indian region except the peak in correlation in

NH is located near 20–25°N than 15–20°N. Since, DCH values are less in this longitude sector of Pacific region (Chun et al., 2007), even though correlations are high. During El Niño year (2009), very weak or almost no correlations are seen over equatorial region and NH, this is unlike other regions of study. Rather the correlations are higher in normal years in NH. On the other hand, during 2008 a La Nina year correlations are higher in the entire region of Western Pacific, particularly over 10°S to 15°N below ABL. Inter-annual variability of correlation coefficient over western Pacific region suggests that El Niño tends to develop negative correlation while La Nina favours positive correlation between CPT and lower tropospheric region.

4. Summary and concluding remarks

Vertical correlations are computed between lower troposphere and CPT region over different regions of convective activities over the globe using RO data derived from COSMIC during 2007–2010. Zonally averaged correlations, which are further averaged over tropical region (from 30°S to 30°N) turns out to be nearly 0.4, which is consistent with the result by Wu et al. (2006). In both hemispheres longitudinally averaged features show a positive correlation in the NH, while negative correlations are quite dominant in SH. Inter-annual variation in latitudinal variability of correlations demonstrates clearly the influence of La Nina (2008) and El Niño (2009) events. Indian region strongly shows positive correlations over 15–20°N, which are consistent with deep convective heating results by Chun et al. (2007). Latitudinal features of correlations variability over African region presented almost a symmetrical structure on both side of equator, which is quite unique over the globe and consistent with the patterns of convectively deep heating distribution. In the years 2008 and 2009, larger correlations are seen, particularly during 2008 in SH there was a significant rise suggesting this as a highly influenced region by La Nina in terms of vertical coupling. Despite the less visible convective heating in the Western Pacific region, there is a good correlation over 20–25°N, with magnitude as good as over Indian region. Interestingly, El Niño has produced no correlation over this region in NH, while La Nina year of 2008 showed an increase in positive correlations over equatorial region as well as towards wider tropical region in NH. In general, over most of the parts of the globe deep convective heating activities significantly link lower

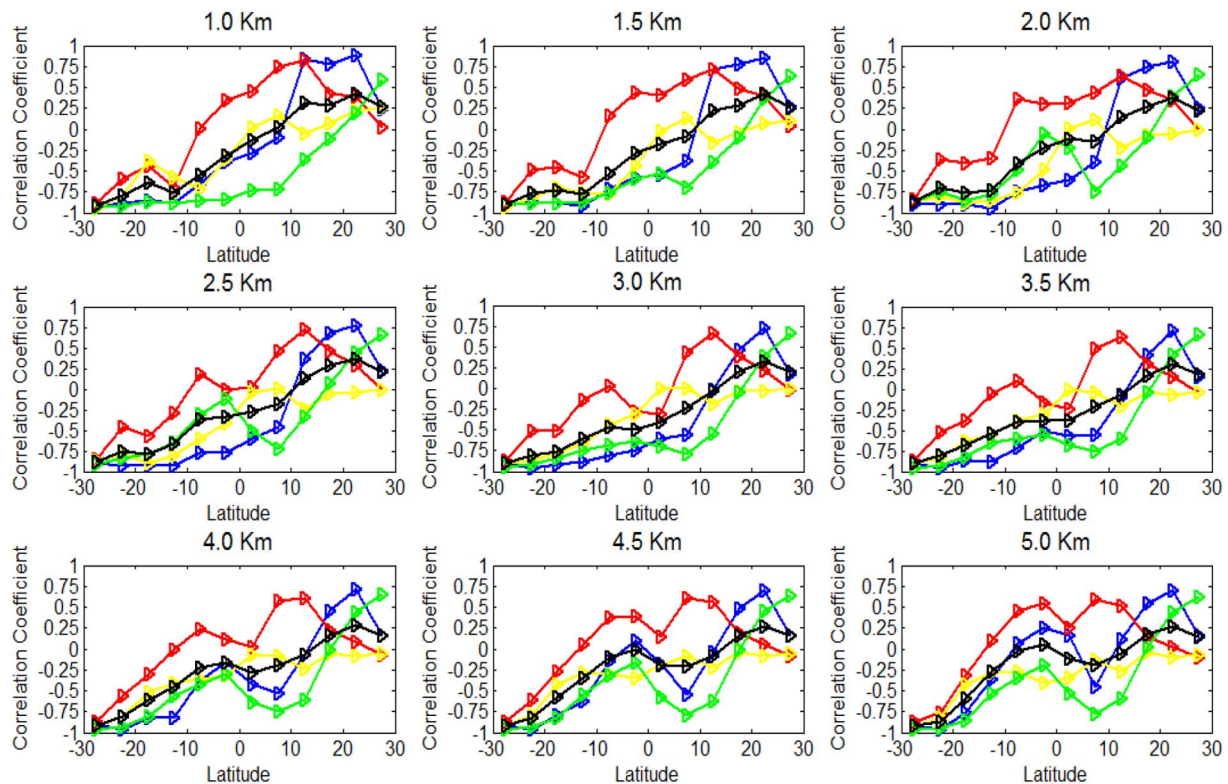


Fig. 6. Same as Fig. 3, but over Western Pacific (120°W–180°W).

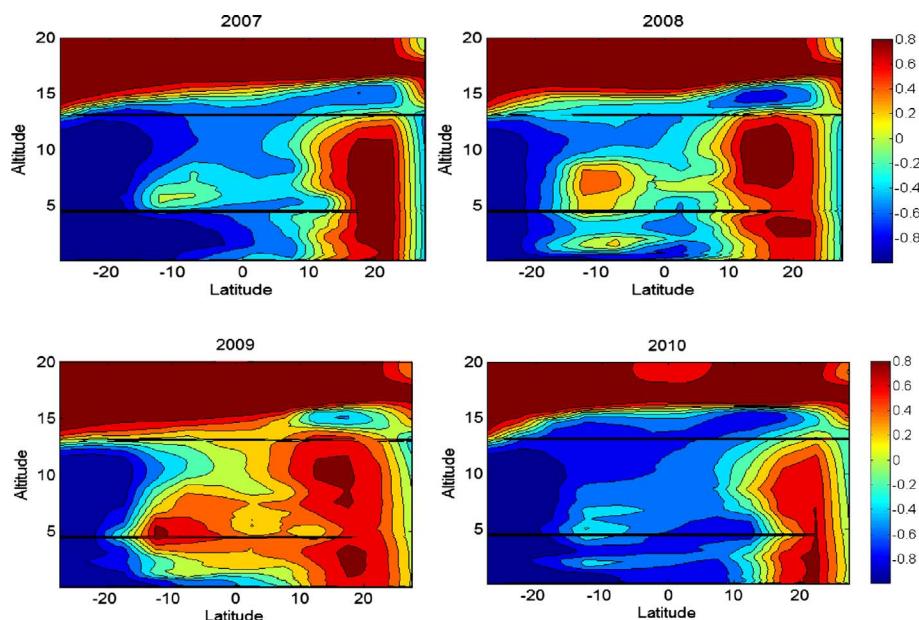
troposphere to CPT region with a significant inter annual variation due to El Niño and La Nina occurrence. Over oceans, in SH particularly, lower and upper troposphere are negatively correlated over less convective region. However, we need to expand the study to explore high values of correlations in Western Pacific region despite the fact of less convective heating.

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Annexure A

Latitude-height sections of vertical correlations between CPT and at each 100 m interval from 1 km to 20 km during 2007–2010. Inter-annual variations of correlations values are quite clear. Particularly, enhancement of positive correlations is seen in Southern Hemisphere during 2009 – an ENSO year. Marginal increase is also noted during 2008 – a La Nina year. As mentioned in the text, the influence of ENSO was quite strong over Indian longitudes region, while La Nina showed a prominent influence over African and Western Pacific region. Between 13 km and 16 km altitudes, there is a negative correlation, which are consistent with the findings of Wu et al. (2006), most probably related with the upwelling in the upper troposphere due to large heating in the middle tropical troposphere, and partly it is related with the lag of 2–3 months in passing the quasi - periodic fluctuations from 200 mb to 100 mb level as discussed by Panwar et al. (2012).



References

- Anthes, R., Rocken, C., Kuo, Y.-H., 2000. Applications of COSMIC to meteorology and climate. *Terr. Atmos. Ocean. Sci.* 11, 115–156.
- Anthes, R.A., Bernhardt, P., Chen, Y., Cucurull, L., Dymond, K., Ector, D., Healy, S., Ho, S., Hunt, D., Kuo, Y., Liu, H., Manning, K., McCormick, C., Meehan, T.K., Randel, W.J., Rocken, C., Schreiner, W.S., Sokolovskiy, S.V., Syndergaard, S., Thompson, D.C., Yen, K.E., Trenberth, N.L., Zeng, Z., 2008. The COSMIC/FORMOSAT-3 mission: early results. *Bull. Am. Meteorol. Soc.* 89, 313–333. <http://dx.doi.org/10.1175/BAMS-89-3-313>.
- Ao, C.O., Meehan, T.K., Hajj, G.A., Mannucci, A.J., Beyerle, G., 2003. Lower troposphere refractivity bias in GPS occultation retrievals. *J. Geophys. Res.* 108 (D18), 4577. <http://dx.doi.org/10.1029/2002JD003216>.
- Bauer, M., Del Genio, A.D., Lanzante, J.R., 2002. Observed and simulated temperature humidity relationships: sensitivity to sampling and analysis. *J. Clim.* 15, 203–215.
- Biondi, R., Randel, W., Ho, S.-P., Neubert, T., Syndergaard, S., 2012. Thermal structure of intense convective clouds derived from GPS radio occultations. *Atmos. Chem. Phys.* <http://dx.doi.org/10.5194/acp-12-5309-2012>.
- Biondi, R., Ho, S.-P., Randel, W., Neubert, T., Syndergaard, S., 2013. Tropical cyclone cloud-top heights and vertical temperature structure detection using GPS radio occultation measurements. *J. Geophys. Res.* 118, 1–13. <http://dx.doi.org/10.1002/jgrd.50448>.
- Chun, H.-Y., Goh, J.-S., Song, I.-S., Ricciardulli, L., 2007. Latitudinal variations of convective source and propagation condition of inertia-gravity waves in the tropics. *J. Atmos. Sci.* 64, 1603–1618.
- Chun, H.-Y., Kim, Y.-H., Choi, H.J., Kim, J.Y., 2011. Influence of gravity waves in the tropical upwelling: WACCM simulations. *J. Atmos. Sci.* 68, 2599–2612. <http://dx.doi.org/10.1175/JAS-D-11-022.1>.
- Dhaka, S.K., Choudhary, R.K., Malik, S., Shibagaki, Y., Yamanaka, M.D., Fukao, S., 2002. Observable signatures of a convectively generated wave field over the tropics using Indian MST radar at Gadanki (13.5°N, 79.2°E). *Geophys. Res. Lett.* 29 (18), 1872–1875.
- Dhaka, S.K., Takahashi, M., Kawatani, Y., Shibagaki, Y., Fukao, S., 2003. Observations of deep convective updrafts in tropical convection and their role in the generation of gravity waves. *J. Meteor. Soc. Jpn.* 81 (5), 1185–1199.
- Dhaka, S.K., Bhatnagar, R., Shibagaki, Y., Fukao, S., Kozu, T., Malik, V., Malik, S., Dutta, A., 2007. Study of temporal variation of equatorial tropopause due to atmospheric waves in CPEA campaign 2004 at Koto Tabang, Indonesia. *Adv. Geosci.* 11, 167–175.
- Dhaka, S.K., Sapra, R., Panwar, V., Goel, A., Bhatnagar, R., Kaur, M., Mandal, T.K., Jain, A.R., Chun, H.-Y., 2010. Influence of large-scale variations in convective available potential energy (CAPE) and solar cycle over temperature in the tropopause region at Delhi (28.3°N, 77.1°E), Kolkata (22.3°N, 88.2°E), Cochin (10°N, 77°E), and Trivandrum (8.5°N, 77.0°E) using radiosonde during 1980–2005. *Earth Planets Space* 62 (2010), 319–331.
- Dhaka, S.K., Kumar, V., Choudhary, R.K., Shu-Peng, Ho, Takahashi, M., Yoden, S., 2015. Indications of a strong dynamical coupling between the polar and tropical regions during the sudden stratospheric warming event January 2009, based on COSMIC/FORMOSAT-3 satellite temperature data. *Atmos. Res.* 166, 60–69. <http://dx.doi.org/10.1016/j.atmosres.2015.06.008>.
- Garcia, R.R., 1987. On the mean meridional circulation of the middle atmosphere. *J. Atmos. Sci.* 44, 3599–3609.
- Gettelman, A., de F. Forster, P.M., 2002. A climatology of the tropical tropopause layer. *J. Meteorol. Soc. Jpn.* 80, 911–942.
- Gray, L.J., Dunkerton, T.J., 1990. The role of the seasonal cycle in the quasi-biennial oscillation of ozone. *J. Atmos. Sci.* 47, 2429–2451.
- Gupta, A., Dhaka, S.K., Panwar, V., Bhatnagar, R., Kumar, V., Datta, Savita M., Dash, S.K., 2013. AIRS observations of seasonal variability in meridional temperature gradient over Indian region at 100 hPa. *J. Earth Syst. Sci.* 122, 201–213.
- Hajj, G.A., Lee, I.C., Pi, X., Romans, L.J., Schreiner, W.S., Straus, P.R., Wang, C., 2000. COSMIC GPS ionospheric sensing and space weather. *Terr. Atmos. Ocean. Sci.* 11 (1), 235–272.
- Haynes, P.H., Marks, C.J., McIntyre, M.E., Shepherd, T.G., Shine, K.P., 1991. On the “downward control” of extratropical diabatic circulations by eddy-induced mean zonal forces. *J. Atmos. Sci.* 48, 651–678.
- Held, I.M., 1982. On the height of the tropopause and the static stability of the troposphere. *J. Atmos. Sci.* 39, 412–417.
- Highwood, E., Hoskins, B., 1998. The tropical tropopause. *Q. J. R. Meteorol. Soc.* 124 (601), 1579–1604.
- Ho, S.-P., Kuo, Y.H., Sokolovskiy, S., 2007. Improvement of the temperature and moisture retrievals in the lower troposphere using AIRS and GPS radio occultation measurements. *J. Atmos. Oceanic Tech.* 1726–1739. <http://dx.doi.org/10.1175/JTECH2071.1>.
- Ho, S.P., et al., 2009. Estimating the uncertainty of using GPS radio occultation data for climate monitoring: intercomparison of CHAMP refractivity climate records from 2002 to 2006 from different data centers. *J. Geophys. Res.-Atmos.* 114 20 pp. <http://dx.doi.org/10.1029/2009JD011969>.
- Ho, S.-P., Zhou, X., Kuo, Y.-H., Hunt, D., Wang, J.-H., 2010a. Global evaluation of radiosonde water vapor systematic biases using GPS radio occultation from COSMIC and ECMWF analysis. *Remote Sens.* 2 (5), 1320–1330 2010.
- Ho, S.-P., Kuo, Ying-Hwa, Schreiner, William, Zhou, Xinjia, 2010b. Using SI-traceable global positioning system radio occultation measurements for climate monitoring. [In *States of the Climate in 2009*]. *Bull. Amer. Meteor. Soc.* 91 (7), S36–S37 (invited).
- Ho, S.-P., Hunt, Doug, Steiner, Andrea K., Mannucci, Anthony J., Kirchengast, Gottfried, Gleisner, Hans, Heise, Stefan, von Engeln, Axel, Marquardt, Christian, Sokolovskiy, Sergey, Schreiner, William, Scherllin-Pirscher, Barbara, Ao, Chi, Wickert, Jens, Syndergaard, Stig, Lauritsen, Kent B., Leroy, Stephen, Kursinski, Emil R., Kuo, Ying-Hwa, Foelsche, Ulrich, Schmidt, Torsten, Gorbunov, Michael, 2012. Reproducibility of GPS Radio Occultation Data for Climate Monitoring: Profile-to-Profile Intercomparison of CHAMP Climate Records 2002 to 2008 from Six Data Centers. *J. Geophys. Res.* 117, D18111. <http://dx.doi.org/10.1029/2012JD017665>.
- Ho, S.-P., Yue, X., Zeng, Z., Ao, C., Huang, C.-Y., Kursinski, E.R., Kuo, Y.-H., 2013. Applications of COSMIC radio occultation data from the troposphere to ionosphere and potential impacts of COSMIC-2 data. *BAMS*. <http://dx.doi.org/10.1175/BAMS-D-13-00035.1>.
- Ho, S.-P., Liang, P., Richard, A., Anthes, R., Kuo, Ying-Hwa, Lin, Hsiao-Chun, 2015. Marine boundary layer heights and their longitudinal, diurnal, and interseasonal variability in the Southeastern Pacific using COSMIC, CALIOP, and radiosonde data. *J. Clim.* 28, 2856–2872. <http://dx.doi.org/10.1175/JCLI-D-14-00238.1>.
- Holton, J.R., Haynes, P.H., McIntyre, M.E., Douglass, A.R., Rood, R.B., Pfister, L., 1995. Stratosphere-troposphere exchange. *Rev. Geophys.* 33, 403–439.
- Huang, Ching-yuang, Wen-hsin, Teng, Ho, S.-P., Kuo, Y.H., 2013. Global variation of COSMIC precipitable water over land: comparisons with ground-based GPS measurements and NCEP reanalyses. *GRL*. <http://dx.doi.org/10.1002/grl.50885>.
- Jensen, E.J., Toon, O.B., Selkirk, H.B., Spinhirne, J.D., Schoeberl, M.R., 1996. On the formation and persistence of subvisibelicirrus clouds near the tropical tropopause. *J. Geophys. Res.* 101 (D), 21 361–21 375.
- Kerr-Munslow, A.M., Norton, W.A., 2006. Tropical wave driving of the annual cycle in

- tropical tropopause temperatures. Part I: ECMWF analyses. *J. Atmos. Sci.* 63, 1410–1419.
- Kim, J., Son, S.-W., 2012. Tropical cold-point tropopause: climatology, seasonal cycle, and intraseasonal variability derived by COSMIC GPS radio occultation measurements. *J. Clim.* 25, 5343–5360. <http://dx.doi.org/10.1175/JCLI-D-11-00554.1>.
- Kishore, P., Namboothiri, S.P., Jiang, J.H., Sivakumar, V., Igarashi, K., 2009. Global temperature estimates in the troposphere and stratosphere: a validation study of COSMIC/FORMOSAT-3 measurements. *Atmos. Chem. Phys.* 9, 897–908.
- Kumar, V., Dhaka, S.K., Jain, A., Chaudhary, A., Bhatnagar, R., Gupta, A., Panwar, V., Singh, N., Reddy, K.K., 2013. COSMIC satellite observations on seasonal variation of pressure at cold point tropopause (P-CPT) and its relation with Tropical Easterly Jet (TEJ) over tropical region. *Indian J. Radio Space Phys.* 42, 292–297.
- Kumar, V., Dhaka, S.K., Choudhary, R.K., Ho, Shu-Peng, Reddy, K.K., 2014. On the occurrence of coldest temperature in stratospheric region and tropical tropopause stability: A study using COSMIC/ FORMOSAT-3 satellite measurements. *J. Atmos. Terr. Phys.* 121 (B), 271–286. <http://dx.doi.org/10.1016/j.jastp.2014.10.007>.
- Kuo, Y.-H., Sokolovskiy, S.V., Anthes, R.A., Vandenberghe, F., 2000. Assimilation of GPS radio occultation data for numerical weather prediction. *Terr. Atmos. Ocean. Sci.* 11 (1), 157–186.
- Kursinski, E.R., Hajj, G.A., Schofield, J.T., Linfield, R.P., Hard, R.R., 1997. Observing Earth's atmosphere with radio occultation measurements using the Global Positioning System. *J. Geophys. Res.* 102, 23429–23465.
- Lanzante, J.R., 1996. Resistant, robust and non-parametric techniques for the analysis of climate data: theory and examples, including applications to historical radiosonde station. *Int. J. Climatol.* 16, 1197–1226.
- Mears, C., Wang, J., Ho, S.-P., Zhang, L., Zhou, X., 2012. Hydrological cycle, total column water vapor. [in “State of the Climate in 2011”]. *Bull. Amer. Meteor. Soc.* 93 (7), S25–S26 [Global Climate]. (Doi: 10.1175).
- Mears, C., Wang, J., Ho, S.-P., Zhang, L., Zhou, X., 2013. Total column water vapor. [In States of the Climate in 2012]. *Bull. Amer. Meteor. Sci.* (in press).
- Olsen, E., et al., 2005. AIRS/AMSU/HSB Version 4.0 Data Release User Guide. 72 pp. Jet. Propul. Lab., Calif. Inst. of Technol., Pasadena (Dec).
- Panwar, V., Jain, A.R., Goel, A., Mandal, T.K., Rao, V.R., Dhaka, S.K., 2012. Some features of water vapor mixing ratio in tropical upper troposphere and lower stratosphere: role of convection. *Atmos. Res.* 108, 86–103. <http://dx.doi.org/10.1016/j.atmosres.2012.02.003>.
- Randel, W.J., Hensen, E.J., 2013. Physical processes in the tropical tropopause layer and their roles in a changing climate. *Nat. Geosci.* 6, 169–176. <http://dx.doi.org/10.1038/ngeo1733>.
- Randel, W.J., Wu, F., Rios, W.R., 2003. Thermal variability of the tropical tropopause region derived from GPS/MET observations. *J. Geophys. Res.* 108. <http://dx.doi.org/10.1029/2002JD002595>.
- Ratnam, M.V., Basha, S.G., 2010. A robust method to determine global distribution of atmospheric boundary layer top from COSMIC GPS RO measurements. *Atmos. Sci. Lett.* 11, 216–222. <http://dx.doi.org/10.1002/asl.277>.
- Rieckh, T., Anthes, R.A., Randel, W., Ho, S.-P., Foelsche, U., 2017. Tropospheric dry layers in the tropical western Pacific: comparisons of GPS radio occultation with multiple data sets. *Atmos. Meas. Tech.* 10, 1093–1110. <http://dx.doi.org/10.5194/amt-10-1093-2017>.
- Scherllin-Pirscher, B., Deser, C., Ho, S.-P., Chou, C., Randel, W., Kuo, Y.-H., 2012. The vertical and spatial structure of ENSO in the upper troposphere and lower stratosphere from GPS radio occultation measurements. *Geophys. Res. Lett.* 39, L20801. <http://dx.doi.org/10.1029/2012GL053071>.
- Sherwood, S.C., 2000. A stratospheric “drain” over the maritime continent. *Geophys. Res. Lett.* 27, 677–680.
- Sherwood, S.C., Horinouchi, T., Zeleznik, H.A., 2003. Convective impact on temperatures observed near the tropical tropopause. *J. Atmos. Sci.* 60, 1847–1856. [http://dx.doi.org/10.1175/1520-0469\(2003\)060<1847:CIOTON>2.0.CO;2](http://dx.doi.org/10.1175/1520-0469(2003)060<1847:CIOTON>2.0.CO;2).
- Sokolovskiy, S.V., 2001. Tracking tropospheric radio occultation signals from low earth orbit. *Radio Sci.* 36, 483–498. <http://dx.doi.org/10.1029/1999RS002305>.
- Sun, D.Z., Oort, A.H., 1995. Humidity-temperature relationships in the tropical troposphere. *J. Clim.* 8, 1974–1987.
- Teng, Wen-Hsin, Ching-Yung, Huang, Ho, S.-P., Kuo, Ying-Hwa, Zhou, Xin-Jia, 2013. Characteristics of global precipitable water in ENSO events revealed by COSMIC measurements. *J. Geophys. Res.* 118, 1–15. <http://dx.doi.org/10.1002/jgrd.50371>.
- Ueyama, R., Wallace, J.M., 2010. To what extent does high-latitude wave forcing drive tropical upwelling in the Brewer–Dobson circulation? *J. Atmos. Sci.* 67, 1232–1246.
- Wu, W., Dessler, A.E., North, G.R., 2006. Analysis of the correlations between atmospheric boundary-layer and free-tropospheric temperatures in the tropics. *Geophys. Res. Lett.* 33, L20707. <http://dx.doi.org/10.1029/2006GL026708>.
- Yulaeva, E., Holton, J.R., Wallace, J.M., 1994. On the cause of the annual cycle in tropical lower-stratospheric temperatures. *J. Atmos. Sci.* 51, 169–174.
- Zeng, Z., Ho, S.-P., Sokolovskiy, S., Kuo, Y.-H., 2012. Structural evolution of the Madden-Julian oscillation from COSMIC radio occultation data. *J. Geophys. Res.-Atmos.* 117 (D22). <http://dx.doi.org/10.1029/2012JD017685>.